

VISIONCLAW AGENT PLATFORM

Comprehensive Feature Document

AI Buddy LLC, Illinois | March 2026

visionclawagent.com

A Self-Actualizing Multi-Tenant Agentic AI Corporation Platform

~89,000 lines of TypeScript | 130+ files | 66 database tables
95 agentic tools | 36+ AI models | 14 AI personas
22 page components | 190+ API endpoints | 50+ skills

Full-stack monolith — React 18 + Express.js + PostgreSQL, single-port (5000)

- Multi-tenant SaaS with isolated workspaces per user
- 14-persona AI agent team with autonomous background tasks
- 6+ AI provider integrations with intelligent routing
- Real-time streaming via Server-Sent Events (SSE)
- Dark/light theme with full mobile PWA support
- Installable as a Progressive Web App on any device

AUTHENTICATION & USER MANAGEMENT

Three sign-up methods — Replit Social Login (Google/Apple/GitHub via OpenID Connect), Email + Password, Admin PIN

- PBKDF2-SHA512 password hashing (100,000 iterations) with timing-safe comparison
- HMAC-SHA256 PIN hashing with timing-safe comparison
- Database-backed sessions (survive server restarts)
- 7-day session tokens stored in localStorage

Auto-provisioning — new users automatically get a tenant on first login

- Account linking prevents duplicate accounts for Replit social login users
- Auth-aware file serving with query parameter token support

MULTI-TENANT ISOLATION

Every piece of data scoped by tenant_id — conversations, memories, knowledge, heartbeat tasks, marketing data, experiments, browser sessions, email inboxes

- IDOR-protected routes verify ownership on all CRUD operations

Plan tiers — Trial (5 conversations), Starter, Pro, Enterprise

- Trial limits with upgrade prompts and banners

Admin vs user separation — admin has full access, non-admin see filtered sidebar

- Per-tenant email inboxes auto-provisioned via AgentMail
- Per-tenant browser contexts with separate cookies/storage
- Per-tenant API key management (BYOK)

- Per-tenant WhatsApp connections
- Per-tenant Coinbase Commerce keys
- Per-tenant file storage
- Per-tenant usage metering

AI CHAT ENGINE

- Unlimited concurrent conversations, each with its own model and persona
- Real-time token streaming via SSE** — tokens appear as generated
- Markdown rendering with syntax highlighting, LaTeX math, and code blocks
- "Think then Act" reasoning mode** — step-by-step thoughts in collapsible blocks
- Conversation management** — create, rename, delete, search

- Auto-titling based on first message
- Abort generation mid-stream with cancel button
- Multi-round tool loops** — AI chains up to 5 tool calls per turn
- Copy to clipboard, view raw content

Smart Context Injection — time-of-day greeting, recent conversations, remembered

user facts

- Context window auto-compaction** — conversations over 20 messages are summarized
- Conversation truncation at 95% context usage (preserves system prompt + recent)

MULTI-PROVIDER AI ROUTING (36+ MODELS)

OPENAI: GPT-5.4, GPT-5.1, GPT-5, GPT-5-Mini, GPT-5-Nano, GPT-4o, o4-mini

ANTHROPIC: Claude Opus 4.6, Claude Sonnet 4.6, Claude Sonnet 4

GOOGLE: Gemini 3 Flash Preview, Gemini 2.5 Pro

xAI: Grok 4.1, Grok-3, Grok-3-mini

PERPLEXITY: Sonar, Sonar Pro, Sonar Deep Research, Sonar Reasoning Pro

OPENROUTER: DeepSeek V3.2, DeepSeek R1, GLM-5, GLM-4.5, Kimi K2.5,

MiniMax M2.5, Qwen 3.5, Llama 4, Mistral, and more

- Dynamic model discovery from each provider's API
- Cost tier badges per model (Free, Low, Medium, High, Premium)
- Per-conversation model selection** — switch models mid-conversation
- Automatic failover if a provider is unavailable
- Provider API key management UI with connection testing

INTELLIGENT AUTO-SELECT ROUTER

Classifies each message across 10 task categories — simple-chat, coding, reasoning, research, vision, agentic, writing, translation, data-analysis, general

- Fast heuristic classification via pattern matching (< 1ms)
 - LLM fallback when heuristics are inconclusive
 - Only routes to models the user has configured and available
 - Route decisions streamed as SSE events, stored in message metadata
 - Amber badges in UI showing the routed model name
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INTELLIGENT MODEL COST ROUTER (HEARTBEAT)

- Automatically assigns cheapest appropriate model to autonomous tasks
- Fast tier (\$)** — Radar, Atlas, Scribe, Chief of Staff -> gpt-5-nano
- Balanced tier (\$\$)** — Apollo, Forge, Proof -> gpt-5-mini
- Powerful tier (\$\$\$)** — VisionClaw, Felix, Neptune -> conversation model
- User-created tasks keep their explicit model selection
 - Heartbeat logs record the effective (routed) model for cost analytics

14-PERSONA AI AGENT SYSTEM

1. VisionClaw — Personal Assistant (default conversational agent)
2. Felix — CEO & Orchestrator (revenue growth, high-leverage execution)
3. Forge — Staff Engineer (code quality, reliability)
4. Teagan — Content Marketing (strategy and copy)
5. Blueprint — Multi-Agent Operator (orchestration and coordination)
6. Chief of Staff — Operations Director (task routing, chain of command)
7. Scribe — Content Creator (blog posts, social media, emails)
8. Proof — Content Reviewer (quality gate, approves/revises Scribe's output)
9. Radar — Intelligence Analyst (daily scans, trend detection)
10. Neptune — Deep Research Specialist (activated on Radar escalation)
11. Apollo — Revenue & Pipeline (sales tracking, deal progression)
12. Atlas — Metrics & Reporting (ROI tracking, cost analysis)
13. Cassandra — Risk & Compliance
14. Luna — Creative & Design

PERSONA FEATURES:

- 8 document fields per persona** — Soul, Identity, Memory Doc, Operating Loop, Heartbeat Doc, Tools Doc, Agents Doc, Brand Voice Doc
- Quick-switch dropdown in chat header
 - Chain of command enforcement (Neptune only via Radar, no direct CEO access)
 - Content two-gate rule (Scribe drafts, Proof must review)
 - Full CRUD for custom personas
 - Paid users can rename personas (tenant-scoped, original preserved)

95 AGENTIC TOOLS

SYSTEM TOOLS (3):

- test_api_keys, check_system_status, list_models

MEMORY & KNOWLEDGE TOOLS (7):

- search_memory, create_memory, update_memory, search_knowledge, create_knowledge, recall_context, get/write daily notes

COMMUNICATION TOOLS (3):

- send_email (AgentMail), check_inbox, list_conversations

WEB & RESEARCH TOOLS (3):

- web_fetch (Jina AI reader), web_search (Wikipedia/Perplexity/Jina), deep_research (multi-stage pipeline with structured reports)

BROWSER TOOL (18 ACTIONS):

- navigate, screenshot, content, click, type, evaluate, smart_browse,

form_fill, open_tab, close_tab, focus_tab, tabs, snapshot, wait, select, pdf, health, close_session

PDF TOOLS (5):

- analyze_pdf, create_pdf, fill_pdf, edit_pdf, list_pdf_fields

CODE & EXECUTION TOOLS (4):

- execute_code (hardened VM sandbox), exec (sandboxed shell), show_diff, llm_task

PLANNING & ORCHESTRATION TOOLS (6):

- plan_and_execute, delegate_task, sessions_spawn, sessions_list, sessions_history, sessions_send, subagents

WORKFLOW TOOLS (1):

- lobster (multi-step workflows with approval gates)

VISUALIZATION (1):

- generate_chart (bar, line, pie, area — inline interactive charts)

FILE & STORAGE TOOLS (5):

- list_uploads, google_drive, deliver_product, delivery_status, project

SELF-IMPROVEMENT TOOLS (6):

- create_tool, list_custom_tools, delete_custom_tool, log_experiment, get_experiments, run_self_improvement

MARKETING TOOLS (4):

- draft_social_post, manage_content_calendar, marketing_analytics, marketing_experiment

SKILLS MANAGEMENT (1):

- manage_skills (50+ toggleable agent capabilities)

ADDITIONAL TOOLS (28+):

- Credential vault, Google Drive operations, calendar & scheduling, WhatsApp messaging, Discord bot, voice TTS/STT, file reading, read_file with multi-directory search, and more

TOOL INTELLIGENCE FEATURES

Multi-round tool loops — chains multiple tool calls (up to 5 rounds/turn)

Self-healing tool loop — failed calls get error analysis hints injected back to the model (up to 2 retries)

Tool loop detection — hash-based tracking prevents infinite loops (generic repeat, ping-pong, poll-no-progress, circuit breaker)

Mutation tracking — classifies every tool call by risk level (read-only, write, destructive)

Tool execution timeouts — 60s default, 120s for slow tools

Failover with tool-support recomputation — provider failure triggers cascade

- Tool results rendered inline with collapsible detail sections in chat

SELF-HEALING & ADAPTIVE PLANNING

Self-Healing Tool Loop — error detection -> structured recovery hints -> retry tracking (max 2) -> SSE events -> UI indicators -> graceful degradation

Adaptive Re-Planning — failed task planner steps trigger dynamic restructuring

-> only affected steps modified -> alternative approaches -> max 2 re-plans
-> step ID continuity -> progress events

AGENT SELF-REFLECTION

4-axis evaluation — accuracy, completeness, relevance, tone (1-10 each)

- Automatic refinement if overall score < 7 or any axis < 5
- Fast evaluation via GPT-5-Nano (< 500ms)
- Refinement via GPT-5-Mini with specific critique
- Max 2 reflection rounds
- SSE streaming with mirror indicator in UI
- Skips for short responses (< 50 chars) and thinking mode

TOOL LEARNING SYSTEM (RUNTIME CUSTOM TOOLS)

Natural language tool creation — describe what it should do, system generates implementation

- LLM-generated implementations with sandboxed execution
- Persistent registry in database (survives restarts)
- Dynamic loading into tool definitions at runtime
- Validation on create with try-and-revert testing (3 LLM-written test cases)
- Auto-fix on failure (up to 2 attempts), revert on exhaustion
- Every creation attempt logged as an experiment
- Auto-prefixed names (custom_) to distinguish from built-in tools

AUTONOMOUS SELF-IMPROVEMENT ENGINE

- Hypothesis generation based on recent reflection scores
- A/B experiment execution (baseline vs experimental response)
- Self-reflection evaluation on both responses
- Head-to-head LLM judge comparison for marginal differences
- Automatic keep/revert (score gain ≥ 0.5 kept, regressions reverted)
- Heartbeat integration** — schedulable via cron
- Categories** — prompt_optimization, response_quality, tool_usage, persona_tuning
- Max 3 experiments per cycle
- Stagnation detection** — auto-skip after 6+ consecutive failures
- Evolution strategies** — balanced, innovate, harden, repair-only
- Signal extraction from server logs (error patterns, tool failures, rate limits)

SUB-AGENT SYSTEM

- Depth-controlled spawning (main -> sub-agent -> sub-sub-agent, max depth 2)
 - Structured Analyze->Execute->Verify->Report task prompts
 - Full tool access (except spawning at max depth)
 - Results reported back to main agent for verification
 - Max 8 concurrent sub-agents, max 5 per parent
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SEMANTIC THREE-TIER MEMORY

DURABLE FACTS — 8 categories — identity, preference, relationship, goal, context, skill, milestone, status

DAILY NOTES — date-scoped logs with events, decisions, lessons, tomorrow's plan

KNOWLEDGE BASE — permanent reference library with title, content, category, priority (1-5)

- LLM-powered fact extraction with confidence scoring on every turn

Contradiction detection — auto-supersedes outdated memories

- Deduplication via cosine similarity + LLM classification
- Semantic search via OpenAI text-embedding-3-small with keyword fallback
- MMR Diversity Re-Ranking (Maximal Marginal Relevance)

Recency tiers — Hot (< 7 days), Warm (7-30 days), Cold (> 30 days) with decay

- Memory Health Monitoring endpoint
- Automated memory snapshots to Google Drive (every 12h, 60-snapshot retention)
- Token budget controls for system prompt injection
- Persona-scoped memories
- Full CRUD management on the Memory page

AUTONOMOUS HEARTBEAT ENGINE

- Cron-based scheduling with 60-second tick loop

10+ supported task types — routine, daily_planning, reflection, memory_consolidation, knowledge, model_scout, delegation, content, content_review, cloud_backup, self_improvement

- Intelligent cost routing per persona tier
- One-click task templates for 6 agents
- Self-task creation and cross-agent delegation
- Dead-letter protection (auto-disable after 5 consecutive failures)
- Exception-safe mutex with try/finally
- Maintenance cycles every 10 ticks
- Activity-based skip to avoid redundant runs

Guardrails — max 10 active delegation tasks, max 60 AI calls/hour, max 5 tasks/tick

- 2-hour delegation expiry

DEEP RESEARCH ENGINE (KARPATHY-INSPIRED)

- Autonomous multi-session research with experiment-based design
- Pre-built research programs with configurable objectives and constraints

"Run All Programs" batch execution — one-click start for all programs

• Automated scheduling with timezone support (Central, Eastern, Mountain, Pacific, UTC)

Preset times — Nightly 2 AM, Midnight, Every Morning 6 AM, Every 6/12 Hours, Weekly, etc.

- Self-evaluating experiments with Keep/Discard/Crash status

Chained results — experiments build on prior findings

- Executive summary generation
- Cost-efficient model routing (DeepSeek V3.2, Qwen 3.5 Flash, R1 Reasoning)
- Dev-to-prod auto-sync on publish

CEO ORCHESTRATOR ("WAR ROOM")

- LLM-powered task decomposition and routing
- Delegates work to specialist personas
- Synthesizes multi-persona responses
- Tracks delegation history and performance

REMOTE BROWSER CONTROL (VIRTUAL BROWSER)

- Per-tenant BrowserContext isolation via Chrome DevTools Protocol
- Multi-profile support** — Browserless, Browserbase, self-hosted Chrome
- Auto-configure from BROWSERLESS_API_KEY
- 18 browser actions including smart_browse, form_fill, close_session
- Screenshots stored per-tenant, auto-pruned after 24h
- Optional base64 inline screenshots for chat context
- SSRF hardening** — IPv4 normalization, IPv6-mapped, DNS fail-closed, cloud metadata blocking, scheme restrictions (http/https only)
- Rate limiting** — max 3 concurrent sessions/tenant, max 30 actions/min/tenant
- Idle session auto-close after 5 minutes
- WebSocket health probe for Browserless endpoints

VOICE CONVERSATIONS

- Text-to-Speech via OpenAI (onyx voice, gpt-4o-mini-tts)** — primary
- ElevenLabs with configurable voice, model, stability, similarity boost
- Microsoft Edge TTS as free fallback
- Speech-to-Text via browser Web Speech API
- Voice Wake and Talk Mode** — hands-free continuous conversation
- Auto mode toggles in Settings
- Configurable max text length and timeout

EMAIL INTEGRATION (AGENTMAIL)

- Per-tenant auto-provisioned @agentmail.to inbox
- Compose, reply, and auto-refresh
- Agent tools** — send_email and check_inbox
- Tenant-scoped routes filter by requesting tenant's inbox
- Full email page UI at /email

WEB SEARCH & CONTENT EXTRACTION

- Perplexity Sonar integration (AI-summarized deep web search)
- Jina AI reader (clean text extraction from URLs)
- Wikipedia search (factual lookups)
- Firecrawl integration (advanced web page extraction and crawling)
- SSRF protection on all web fetching
- Configurable search provider and model

LOBSTER WORKFLOW ENGINE

- YAML-based multi-step workflow definitions
- Step-by-step execution with approval gates for user review
- Security hardening** — path traversal protection, token validation
- Settings UI with workflow list, create/delete, pending approval count

STRIPE PAYMENTS

- Product catalog with one-time or subscription pricing
- 3 subscription tiers** — Starter (\$29/mo), Pro (\$99/mo), Enterprise (\$299/mo)
- Stripe Checkout for seamless payment
- Transaction history with status tracking
- Webhook-driven sync

DUAL TENANT PAYMENT MODES:

Managed (Stripe Connect) — \$99 setup fee, 3% platform commission

BYOK (Bring Your Own Key) — \$29 setup fee, no ongoing fee, tenant's own

Stripe keys (AES-256-GCM encrypted)

COINBASE CRYPTO PAYMENTS

Dual-mode — CDP wallet-based payments + Commerce checkout

CDP Wallet — create EVM wallets, list accounts, check token balances

Commerce — create charges, list transactions, webhook-verified

- HMAC-SHA256 webhook verification

THREE-TIER AUTOMATED BACKUP

Tier 1 — Google Drive Full Backup (daily 3 AM) — all 17 tables as JSON, 30-backup retention, sensitive data auto-redacted

Tier 2 — Memory Snapshots (every 12h) — active + superseded memories, 60-snapshot retention

Tier 3 — GitHub Code Push (after daily backup) — auto-commit and force-push

- Manual triggers available via API endpoints

ANALYTICS DASHBOARD

- Messages per day (area chart)
- Model usage breakdown (pie chart)
- Hourly activity patterns (bar chart)
- Tool usage rankings (bar chart)
- KPI cards** — total conversations, messages, period summaries

SOCIAL MARKETING SUITE

draft_social_post — platform-optimized drafting (X, LinkedIn, TikTok, Instagram)

- manage_content_calendar** — scheduling, campaign grouping, lifecycle
- marketing_analytics** — performance analytics
- marketing_experiment** — A/B experiment tracking with automatic winner detection
 - Brand voice configured per tenant

PUBLIC CHAT & EMBEDDABLE WIDGET

- Shareable link** — /public-chat/:token — standalone dark-themed chat with SSE
- Embeddable widget** — one-line <script> tag adds floating chat bubble to any site
 - Settings toggle with unique 16-character token generation
- Scope isolation** — public conversations tagged separately
- Rate limiting** — 30 req/min general, 10 msg/min streaming
- Safety** — max 2,000 char messages, max 3 tool rounds, only safe tools
- Cost control** — always uses DeepSeek V3.2 via OpenRouter
- Anti-hijacking** — 7-rule security constraint block in system prompt

VANITY SLUG URLs

- Clean branded URLs** — /c/my-company instead of random tokens
 - Settings UI with live preview
- Validation** — 3-40 chars, lowercase alphanumeric + hyphens
 - Reserved word blocking
 - Global uniqueness with database constraint
 - Paid-only feature

PROVIDER KEY PROXY / MASKING SYSTEM

- Two-tier architecture** — Platform keys (master) + Tenant BYOK keys
- Resolution order** — Tenant BYOK -> Platform master -> Replit integration -> Error
- Key masking** — API never returns full values (sk-...xxxx format)
- Health tracking** — last_verified_at, consecutive_failures, last_error
 - AES-256-GCM encryption for stored keys

SKILLS SYSTEM

- 50+ toggleable agent capabilities organized by category
- Skills page at /skills with search and category filtering
- Skills inject additional context and instructions into agent system prompts
- Specialized skills** — Vibe Marketing, Browser Automation, Caption Generation, Coding Agent Loops, and more

INLINE CHART GENERATION

- Bar, line, pie, and area chart types
 - Generated via generate_chart tool, rendered inline in chat
 - Interactive Recharts components with tooltips and legends
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DIGITAL PRODUCT DELIVERY

- Google Drive + email delivery pipeline
- Delivery tracking and status endpoints
- File manifest management

DAILY BRIEFING & WEATHER

- AI-powered personalized daily briefings with weather
- Text-to-Speech option via ElevenLabs
- IP geolocation automatic location detection

DISCORD BOT INTEGRATION

- Bot token configured in Settings
- Responds to Discord messages using active persona and model
- Shares same processMessage engine as chat (full tool access)

SECURITY HARDENING

- Multi-tenant IDOR protection on all CRUD operations
- LLM cost enforcement** — server-side model validation
- Public chat anti-hijacking (7-rule constraint block)
- Public chat rate limiting (30 req/min, 10 msg/min)
- Tool loop detection (hash-based, 4 detection modes)
- Mutation tracking (read-only, write, destructive classification)
- SSRF protection on web_fetch, web_search, and browser tool
- Browser SSRF** — IPv4 normalization, IPv6-mapped, DNS fail-closed, metadata block
- Path traversal protection (basename-only, realpath checks)
- Token validation (strict regex)
- Context window guard (85%/95% threshold auto-truncation)
- External content security (sanitization, 13 suspicious pattern detection)
- Code sandbox isolation (VM context, blocked patterns, frozen objects, 5s timeout)
- Sub-agent depth limits (max depth 2, max 8 concurrent, max 5 per parent)
- Sandboxed shell execution
- HMAC-SHA256 PIN hashing with timing-safe comparison
- PBKDF2-SHA512 password hashing (100k iterations)
- AES-256-GCM encryption for API keys
- DB-backed sessions (not in-memory)
- Secret scanner runs before every GitHub push
- Content Security Policy via Helmet

RESILIENCE & RECOVERY

- Multi-round model failover across providers
- Tool execution timeouts (60s default, 120s for slow tools)
- Self-healing tool retries with error analysis
- Heartbeat dead-letter protection (auto-disable after 5 failures)

- Exception-safe heartbeat mutex (try/finally)
 - Adaptive re-planning (up to 2 re-plans)
 - Context window auto-compaction via summarization
 - Conversation truncation at 95% context usage
- Health Monitor** — automated checks for 6+ core services with auto-remediation

TECH STACK

FRONTEND:

- React 18 + TypeScript
- TailwindCSS + shadcn/ui component library
- Wouter (client-side routing)
- TanStack Query v5 (server state)
- Recharts (data visualization)
- ReactMarkdown (rendering)
- Lucide React + react-icons (icons)

BACKEND:

- Express.js + TypeScript
- Drizzle ORM + PostgreSQL
- Zod (request validation)
- SSE (real-time streaming)
- cron-parser v5 (heartbeat scheduling)
- AgentMail SDK (email)
- Puppeteer-core (browser control)
- js-yaml (workflow parsing)

INFRASTRUCTURE:

- PostgreSQL (primary database, 66 tables)
- Replit AI Integrations (OpenAI, Anthropic, Google)
- Replit Connectors (ElevenLabs, Stripe, Google Drive)
- Vite (frontend build)
- Replit Autoscale deployment (8 vCPU / 16 GiB RAM / 8 Max)

EXTERNAL SERVICES:

- OpenAI, Anthropic, xAI, Google Gemini, Perplexity, OpenRouter (AI providers)
- **Stripe (payments)** — managed + BYOK
- Coinbase CDP + Commerce (crypto payments)
- ElevenLabs (TTS/STT)
- Google Drive (cloud backups)
- AgentMail (email)
- Firecrawl (web extraction)
- Jina AI (URL content extraction)
- Perplexity Sonar (web search)
- Browserless (remote browser)

PAGES & NAVIGATION (22 PAGES)

1. Dashboard (/) — stats, templates, activity feed, quick start
2. Chat (/chat/:id) — streaming conversations with tool calls
3. Memory (/memory) — 3-tier memory management UI
4. Knowledge (/knowledge) — permanent knowledge base
5. Analytics (/analytics) — interactive charts and KPIs

6. Skills (/skills) — 50+ toggleable capabilities
7. Email (/email) — AgentMail inbox
8. Marketing (/marketing) — content calendar and analytics
9. Browser (/browser) — virtual browser interface
10. Heartbeat (/heartbeat) — autonomous task management
11. Deep Research (/research) — programs, sessions, experiments, schedules
12. WhatsApp (/whatsapp) — WhatsApp Web integration
13. WhatsApp Approvals (/whatsapp-approvals) — approval management
14. Wallet (/wallet) — Coinbase crypto management
15. Settings (/settings) — full platform configuration
16. Uploads (/uploads) — file management
17. Projects (/projects) — project management
18. Pricing (/pricing) — subscription plans
19. Terms (/terms) — Terms of Service
20. Privacy (/privacy) — Privacy Policy
21. Public Chat (/public-chat/:token) — shareable chat
22. Landing Page — hero, agent showcase, live stats, pricing

PLATFORM STATISTICS

- ~89,000 lines of TypeScript
- 130+ server-side files
- 66 database tables
- 95 agentic tools
- 36+ AI models across 6+ providers
- 14 AI personas
- 22 page components
- 190+ API endpoints
- 50+ toggleable skills
- 3 subscription tiers + crypto payments

DOMAINS & DEPLOYMENT

Primary — visionclawagent.com
Replit — openclaw-agent.replit.app
Repository — github.com/Huskyauto/VisionClaw-Agent

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